Linear Algebra Formula Sheet

Ayesha Ejaz 22i-0899

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1 Linear Equations

1.9 Linear Transformations

$T(x) = A_{n\times m}x$ is a transformation mapping $T : \mathbb{R}^m \mapsto \mathbb{R}^n$.

A function $T(x) = Ax$ is one-to-one if every column of A is a pivot column, and it is onto if each row is a pivot row.

$Range(T) = Col(A)$, and $Ker(T) = Nul(A)$.

(Assuming A is a square matrix.) To find the inverse of a transformation, T^{-1}, find A^{-1} by row-reducing:

$$\begin{aligned} &\left[A : I\right] \\ &= \left[I : A^{-1}\right] \end{aligned}$$

1.9.1 Reflection

About y-axis:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

About x-axis:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

About y=x:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

1.9.2 Rotation

$$A = \begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix}$$

1.9.3 Composition

The composition of two linear transformations $T : \mathbb{R}^n \mapsto \mathbb{R}^m$ and $S : \mathbb{R}^p \mapsto \mathbb{R}^q$ is equal to $T \circ S : \mathbb{R}^p \mapsto \mathbb{R}^m$ where $n = q$. The domain of the composite function is $\mathbb{R}^p$, and the co-domain is $\mathbb{R}^m$

3 Determinants

- If $det(A) = 0$, then A is not invertible.
- To find the determinant of a matrix $A_{n\times n}$, row reduce it until it is a triangular matrix, and multiply the diagonal entries.
- $$\begin{aligned} det(AB) &= det(A) \times det(B) \\ det(A^2) &= (det(A))^2 \\ det(AA^{-1}) &= det(I) = 1 \end{aligned}$$

- $det(A)$ = area/volume of object formed by the columns of A. If $det(A) = 0$, then the area/volume = 0.

3.1 Co-Factor

$$C_{nm} = (-1)^{n+m} det(M_{nm})$$

3.2 Adjoint

To find the adjoint of of a matrix A, find the co-factor matrix, C, of A. $adj(A) = C^T$.

4 Vector Spaces

A matrix $A_{n \times n}$ is linearly independent if it has n pivot columns.

4.2 Nullspace, Columnspace

$Col(A) + Nul(A)$ = total number of vectors in A.

4.2.1 Columnspace

To find $Col(A)$, row reduce the augmented matrix $Ax = X$:

$$\begin{bmatrix} v_{11} & v_{21} & \dots & v_{p1} & : & x_1 \\ 0 & v_{22} & \dots & v_{p2} & : & x_2 + ax_1 \\ \vdots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \dots & k & : & x_p + \dots + cx_2 + bx_1 \end{bmatrix}$$

$x_p + \dots + cx_2 + bx_1 = k$ is $Col(A)$.

Basis of $Col(A)$ aka the basis of A = the span of the linearly independent vectors of A i.e. the pivot-column vectors of A.

If a vector v exists in $Col(A)$, then the system $Ax = v$ must be consistent.

4.2.2 Nullspace

To find $Nul(A)$, row reduce the augmented matrix $Ax = 0$:

$$\begin{bmatrix} av_{11} & bv_{21} & \dots & cv_{p1} & : & 0 \\ 0 & dv_{22} & \dots & ev_{p2} & : & 0 \\ \vdots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \dots & k & : & 0 \end{bmatrix}$$

$ax_1 + bx_2 + \dots cx_p = 0$ is $Nul(A)$.

Basis of $Nul(A)$ = the span of the linearly independent vectors that form the subspace spanned by $ax_1 + bx_2 + \dots cx_p = 0$.

The nullity of a matrix A is the number of vectors in i.e. the dimension of $Nul(A)$.

If a vector v exists in $Nul(A)$, then $Av = 0$:

4.4 Coordinate Systems

Let a basis $B = \begin{bmatrix} b_1 & b_2 & \dots & b_p \end{bmatrix}$, and a vector v. To find the coordinate vector $|v|_B$, solve the augmented matrix $Bx = v$.

To write the transformation matrices of $T : \mathbb{R}^n \mapsto \mathbb{M}_{pxq}$, apply the transformation on the basis of $\mathbb{R}^n$, and write the answer in the linear combination of the basis of $\mathbb{M}_{pxq}$. The same steps will apply on other transformations e.g. $T : \mathbb{P}_m \mapsto \mathbb{M}_{pxq}$, $T : \mathbb{P}_m \mapsto \mathbb{R}^n$, etc.

4.5 Dimension of a Vector Space

Dimension of A = number of vectors in the basis of A.

4.6 Rank

Rank of A = number of vectors in i.e. dimension of $Col(A)$.

4.7 Change of Basis

Let $B = \{b_1, b_2, \dots, b_p\}$ and $C = \{c_1, c_2, \dots, c_p\}$ be two basis for a vector space V.

Projection Matrix:

$$[\ c_1\ c_2\ \dots\ c_p : b_1\ b_2\ \dots\ b_p\ ]$$
$$[\ I : P_{C \longleftarrow B}\ ]$$

Let x be a vector belonging to the span of B. To find the coordinate vector $[x]_C$ relative to the basis C:

$$\left(P_{C \longleftarrow B}\right)(x) = \left([x]_C\right)$$

*A coordinate vector is always a column vector.

5 Eigenvalues and Eigenvectors

5.1 Characteristic Equation

- To find eigenvalues, solve $det(A - \lambda I) = 0$. To find eigenvectors corresponding to each eigenvalue, substitute eigenvalues into $A - \lambda I$ and solve the linear system.
- Product of eigenvalues = $det(A)$, and sum of eigenvalues = sum of diagonal entries of A.

5.2 Diagonalisation

A matrix $A_{p\times p}$ is diagonalisable iff it has p linearly independent eigenvectors.

$$A^n = PD^nP^{-1}$$

where P is the matrix of eigenvectors, and D is the matrix of eigenvalues of A.

5.3 System of Differential Equations

To solve system of differential equations $X' = AX$:

$$\begin{aligned} X' &= (PDP^{-1})X \\ X &= c_1e^{\lambda_1 t}v_1 + c_2e^{\lambda_2 t}v_2 + \ldots + c_pe^{\lambda_p t}v_p \end{aligned}$$

5.4 Iterative/Difference Equations

The eigenvalues of $ax_k = bx_{k-1} + cx_{k-2}$ with initial value $= x_0$ are found by $a\lambda^2 + b\lambda - c = 0$

5.4.1 $\lambda_1 = \lambda_2$

$$x_k = c_1\lambda^k + c_2k\lambda^k$$

5.4.2 $\lambda_1 \neq \lambda_2$

$$x_k = c_1\lambda_1^k + c_2\lambda_2^k$$

To solve system of difference equations with the initial condition X_k:

$$\begin{aligned} X_n &= A^{n-k}X_k \\ &= (PD^{n-k}P^{-1})X_k \\ &= c_1\lambda_1^{n-k}v_1 + c_2\lambda_2^{n-k}v_2 + \ldots + c_p\lambda_p^{n-k}v_p \end{aligned}$$

If $|\lambda| < 1$ then the system is stable, since the outcome if finite. If $|\lambda| > 1$ then the system is unstable, since the outcome is infinite.

6 Orthogonality and Least Squares

6.1 Orthogonal Matrices

- A matrix A is orthogonal if its columns form an orthonormal set.
- $$A^T = A^{-1}$$
  $$A^TA = AA^T = I$$

6.2 Orthogonal Complement

Let $W = \{v_1, v_2, \ldots, v_p\}$. $W^\perp$ is the set of vectors perpendicular to W. To find $W^\perp$, find $Nul(W)$ in the form $c_1x_1 + c_2x_2 + \ldots + c_px_p = 0$. Span of $\{c_1, c_2, \ldots, c_p\}$ is the $W^\perp$.

6.3 Gram Schmidt Process

Let $\vec{a}$ be the vector whose projection is to be taken on the along $B = \{u_1, u_2, \ldots, u_p\}$.

$$\begin{aligned} Proj_{\text{cols}(B)}\vec{a} &= \frac{\text{cols}(B)\text{cols}(B)^T}{\text{cols}(B)^T \cdot \text{cols}(B)}\vec{a} \\ &= \frac{\vec{a} \cdot \text{cols}(B)}{\text{cols}(B)^T \cdot \text{cols}(B)}\hat{\text{cols}(B)} \end{aligned}$$

Projection matrix:

$$P = \frac{\text{cols}(B)\text{cols}(B)^T}{\text{cols}(B)^T \cdot \text{cols}(B)}$$

Before applying Gram Schmidt, make the basis orthogonal:

$$\begin{aligned} v_1 &= u_1 \\ v_2 &= u_2 - (Proj_{u_2} v_1) \\ &\vdots \\ v_p &= u_p - (Proj_{u_p} v_1 + Proj_{u_p} v_2 + \ldots + Proj_{u_p} v_{p-1}) \end{aligned}$$

Then proceed to take the projection of $\vec{a}$ along $\{v_1, v_2, \ldots, v_p\}$.

6.4 QR Factorisation

$$A = QR$$

where the columns of A are orthogonal, Q is an orthonormal matrix ($Q = \left(\frac{a_1}{\|a_1\|}, \frac{a_2}{\|a_2\|}, \ldots, \frac{a_p}{\|a_p\|}\right)$, and R is a triangular matrix ($R = Q^T A$)

6.5 Householder Transformation

$Q = I - 2vv^T$ where I is the identity matrix, and v is a unit vector.

6.6 Least-Squares

If $Ax = b$ is inconsistent, then we find the least square solution:

$$A\hat{x} = \hat{b}$$

where $\hat{b} = Proj_{col(A)} b$.

$$A^T A\hat{x} = A^T b$$

Least-square error:

$$\|A\hat{x} - b\|$$

6.7 Curve-Fitting

$$A(A^TA)^{-1}(A^Tb) = \hat{b}$$
$$A\hat{x} = \hat{b}$$

6.8 Inner Product Spaces

- $\langle u, v\rangle = u \cdot v$
- p-Norm:
$$\|x\|^p = \sqrt[p]{|x_1|^p + |x_2|^p + \ldots + |x_n|^p}$$

7 Symmetric Matrices and Quadratic Forms

7.1 Orthogonal Diagonalisation

A symmetric matrix is where $A^T = A$

$$A = PDP^{-1} = QDQ^T$$

where P is orthonormal matrix of eigenvectors, and Q is an orthogonal matrix of eigenvalues (i.e. $Q^T = Q^{-1}$).

7.2 Quadratic Forms

$$\begin{aligned} ax^2 + by^2 + cxy \quad &+ dx + ey \quad &+f = 0 \\ x^TAx \quad &+ b^Tx \quad &+c = 0 \end{aligned}$$

To write a given quadratic form $F(x)$ as x^TAx, diagonalise A

$$x = Qy$$
$$A = QDQ^T$$
$$x^TAx + b^Tx + c = y^TDy + b^TQy + c$$

7.3 Positive Definite Matrices

If $x^TAx \geq 0$, then A is positive definite matrices i.e.:

- $\lambda_i \geq 0, \forall i$
- It is linearly independent
- $\det(A_j) \geq 0, \forall j$

7.4 Constrained Optimisation

Given that $F(x) = x^TAx$ and $\|x\|^2 = x^Tx = 1$, the range of the quadratic form $F(x)$:

$$\lambda_{\min} \leq F(x) \leq \lambda_{\max}$$

where λ is an eigenvalue of A.

7.5 Gram Matrix

- $A^T A$ is always a square symmetric whose eigenvalues ≥ 0.
- $A^T A = AA^T = I$ iff A is an orthogonal square matrix whose columns form an orthonormal set.

7.6 Singular Value Decomposition

- The length of Ax (where $x = \{v_1, v_2, \ldots, v_p\}$) i.e. $\|Ax\|^2$ is maximum when λ_i of $A^T A$ has the largest magnitude.
- $A_{\mathrm{n\times m}} = U_{\mathrm{n\times n}} \Sigma_{\mathrm{n\times m}} V^T_{\mathrm{m\times m}}$

Where U is an orthogonal matrix (so $UU^T = U^T U = I$), V is an orthonormal matrix of eigenvectors of $A^T A$ (so $VV^T = V^T V = I$), and Σ are the eigenvectors of $A^T A$ ($\Sigma = \sqrt{D}$). To find U:

7.6.1 n = m

$$u_i = \frac{1}{\sigma_i} A v_i, \forall i$$

7.6.2 n < m

$$u_i = \frac{1}{\sigma_i} A v_i, \forall i, n = m$$
$$u_j = 0, \forall j, n < m$$

7.6.3 n > m

$$A v_i = \sigma_i u_i, \forall n = m$$

Let u_{n+1} be a vector that is orthogonal to $u_1, \ldots, u_n$ (use dot product):

$$v_{n+1} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Then apply Gram Schmidt Process to make the vectors formed from u_{n+1} orthogonal to each other.

8 The Geometry of Vector Spaces

8.1 Affine Hull/Span

The affine hull or affine span of a set of vectors $\{v_1, \ldots v_n\} = c_1 v_1 + \ldots + c_n v_n$ where $c_1 + \ldots + c_n = 1$

8.2 Affine Combinations

$$c_1v_1 + c_2v_2 + \ldots + c_pv_p$$
$$c_1 + c_2 + \ldots + c_p = 1$$
$$c_1, c_2, \ldots, c_p \in \mathbb{R}$$

A point $y \in \mathbb{R}^n$ is an affine combination of $v_1, v_2, \ldots, v_p$ iff $y - v_1$ is a linear combination of $v_2 - v_1, \ldots, v_p - v_1$.

A flat is a translate of a set/subspace S in $\mathbb{R}^n$ by a vector p.

8.3 Affine Dependence

$\{v_1, v_2, \ldots, v_p\}$ is affinely dependent iff $\exists\ c_1, c_2, \ldots, c_3 \in \mathbb{R}$ such that:

$$c_1 + c_2 + \ldots + c_p = 0$$
$$c_1v_1 + c_2v_2 + \ldots + c_pv_p = 0$$

i.e. the set $\{v_2 - v_1, \ldots, v_p - v_1\}$ is linearly dependent.

If $\{v_1, v_2, \ldots, v_p\}$ is an affinely independent set, then the coefficients $c_1, c_2, \ldots, c_p$ are called barycentric coordinates.

8.4 Convex Hull/Span

The convex hull or convex span of a set of vectors $\{v_1, \ldots v_n\} = c_1v_1 + \ldots + c_nv_n$ where $0 \leq c_1, \ldots, c_n \leq 1$.

The convex hull consists of all points lying on line segments between distinct pairs of vectors in the set.

8.5 Convex Combinations

Let $\{v_1, v_2, \ldots, v_p\} \in \mathbb{R}^n$, then $c_1v_1 + c_2v_2 + \ldots + c_pv_p$ is called a convex combination if:

$$c_1 + c_2 + \ldots + c_p = 1$$
$$c_i \geq 0, \forall i$$

9 Bilinear Form

- $\phi : V \times V \mapsto F$
- Let $\phi(u, v)$ be any bilinear transformation with basis $B = \mathbb{R}^n$.

$$\phi(u, v) = u^T Av$$
$$\phi(u, v) = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ \vdots & \vdots \\ a_{n1} & a_{n2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$
$$= u_1(av_1 + bv_2) + u_2(cv_1 + dv_2)$$

- e.g.: let a transformation $\phi(p,q) = \int_0^1 p(x)q(x)dx$ is being applied using non-standard basis, $B = \{b_1, b_2, b_3\}$. First apply ϕ on B, and then apply the transformation on each entry of $\left[\phi\right]_B$:

$$A = \left[\phi\right]_B = \begin{bmatrix} \phi(B_1,B_1) & \phi(B_1,B_2) & \phi(B_1,B_3) \\ \phi(B_2,B_1) & \phi(B_2,B_2) & \phi(B_2,B_3) \\ \phi(B_3,B_1) & \phi(B_3,B_2) & \phi(B_3,B_3) \end{bmatrix}$$

$$\phi(B_1,B_1) = \int_0^1 B_1B_1dx$$

$$\vdots$$

$$\phi(B_3,B_3) = \int_0^1 B_3B_3dx$$

$$[\phi]_B = \begin{bmatrix} \int_0^1 B_1B_1dx & \int_0^1 B_1B_2dx & \int_0^1 B_1B_3dx \\ \int_0^1 B_2B_1dx & \int_0^1 B_2B_2dx & \int_0^1 B_2B_3dx \\ \int_0^1 B_3B_1dx & \int_0^1 B_3B_2dx & \int_0^1 B_3B_3dx \end{bmatrix}$$

- To show that a bilinear transformation, $\phi : \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R}$, is symmetric, first show that it is linear by proving $\phi(c_1u_1 + c_2u_2 + \ldots + c_pu_p, d_1v_1 + d_2v_2 + \ldots + d_qv_q) = c_1d_1\phi(u_1,v_1) + \ldots + c_pd_q\phi(u_p,v_q)$.

9.1 Alternating k-Linear Form/Tensor

- When a bilinear transformation $\phi(u,v) = -\phi(v,u)$, it is called an alternating k-tensor.
- To show that a bilinear transformation is an alternating k-tensor, first show that the transformation is linear, then show that the above condition holds true.

9.2 Tensor Product

- Let v and w be two vectors from $\mathbb{R}^n$. By taking their tensor product, we generate a larger vector subspace (from an existing, smaller vector space).

$$v \otimes w = \begin{pmatrix} v_1w_1 & v_1w_2 & \ldots & v_1w_n \\ v_2w_1 & v_2w_2 & \ldots & v_2w_n \\ \vdots & \vdots & & \vdots \\ v_nw_1 & v_nw_2 & \ldots & v_nw_n \end{pmatrix}$$

- Let $\alpha, \beta : \mathbb{P}_2 \mapsto \mathbb{R}$

$$\begin{aligned} \alpha \otimes \beta &= (\alpha \otimes \beta)(a_1 + b_1x + c_1x^2, a_2 + b_2x + c_2x^2) \\ &= \alpha(a_1 + b_1x + c_1x^2) \cdot \beta(a_2 + b_2x + c_2x^2) \end{aligned}$$

9.3 Wedge Product

Let $f \wedge g \in A_2$ i.e. $f \wedge g(u,v) = f(u)g(v) - f(v)g(u)$

9.4 Symmetric Tensor

When a bilinear transformation $\phi(u, v) = \phi(v, u)$, it is called a symmetric tensor.

Let $f \in A_1$, and $g \in A_2$ where g is a symmetric tensor. Then $f \wedge g(u, v, w) = 0$